EAST WEST UNIVERSITY

Department of Computer Science and Engineering

BSc in Computer Science and Engineering

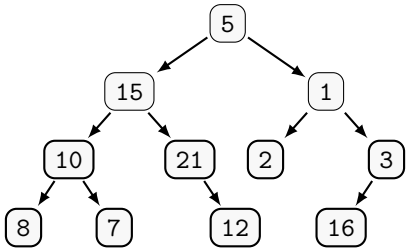
Final Examination, Fall 2025

Course: CSE207 Data Structures, Section-6 **Date:** December 22, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 30
Time: 1 Hour and 20 Minutes

Note: There are **SIX** questions. Answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

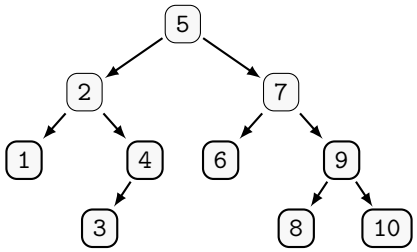
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1. Write down a function for recursive **in-order** traversal of a binary tree. Show the visited nodes following **in-order** traversal for the tree given below.



[CO2, C3, Mark: 5 (4+1)]

2. Determine if the BST given below is a valid AVL tree by computing the height and balance factor for every node.



[CO3, C3, Mark: 5]

3. The following pseudocode is intended to build a **Max-Heap** from an input array $A[1 \ldots n]$. However, the code may contain **logical errors**.

(a) **Identify and explain any possible errors** in the pseudocode.

(b) **Rewrite the corrected pseudocode** for building a Max-Heap.

BUILD-MAX-HEAP(A)
1. $n \leftarrow \text{length}(A)$
2. **for** $i \leftarrow 1$ **to** $\lfloor n/2 \rfloor$
3. **MAX-HEAPIFY**(A, i, n)
4. **end for**

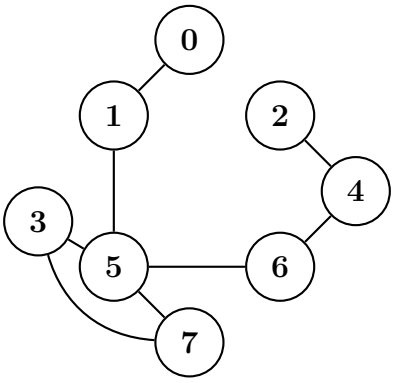
Assume the array is 1-indexed and MAX-HEAPIFY is correctly defined.

[CO3, C3, Mark: 5]

4.

(a) Write the **function/ pseudocode** for **Depth-First Traversal (DFS)** in a graph.

(b) Show the **DFS visit order** of the graph given below.



[CO2, C2, Mark: 5 (3+2)]

5. Perform **Breadth First Traversal (BFS)** of the graph in **Question 4** to find the **minimum distance from source node 0** to all other nodes. You must show:

(a) The **BFS Queue status**

(b) The **Parent node** information for each node

(c) The **Distance** of each node from the source

(d) The final **BFS visit order**

Instruction: You may **cross out outdated/removed values** in your data structures; there is no need to rewrite them repeatedly.

[CO2, C3, Mark: 5]

6.

(a) Explain the role of hash code generation and compression function during hashing.

(b) Describe two methods of collision resolution in hashing.

[CO2, C3, Mark: 5]